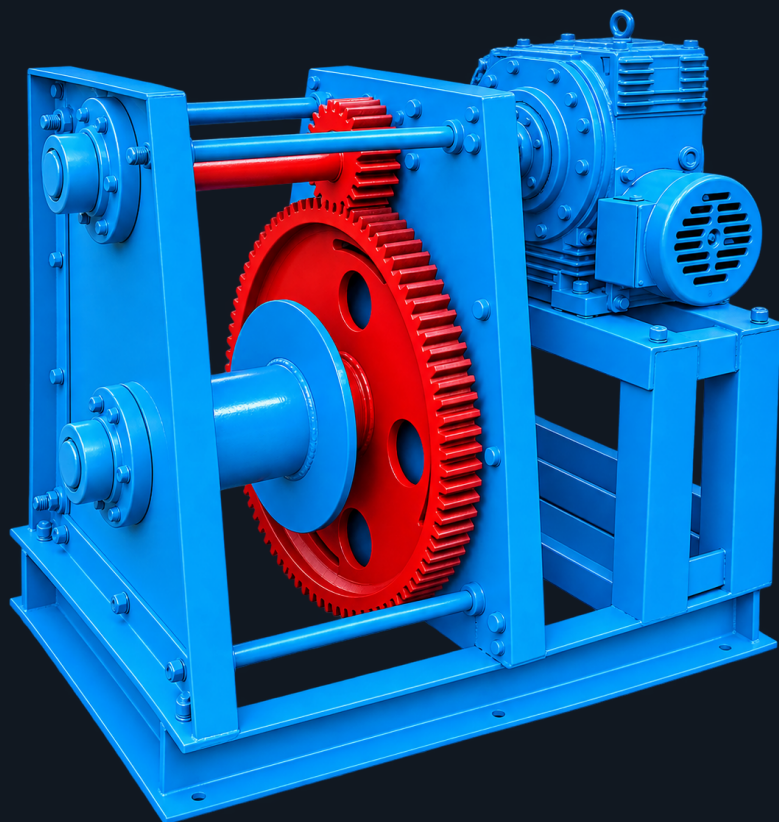




**MANUAL TOOLS COMPANY**

Manufacturer of Coke Oven Machinery

PRODUCT BROCHURE



# DOOR LIFTING POWER WINCH

Coke Oven Gate Lifting Equipment

**2.5 - 5 T**

LIFTING CAPACITY

**5 - 7.5 HP**

MOTOR

**2 - 4 m/min**

LIFTING SPEED

**30+** YEARS OF  
EXCELLENCE

**Ravindra Kr. Agarwal, Proprietor**

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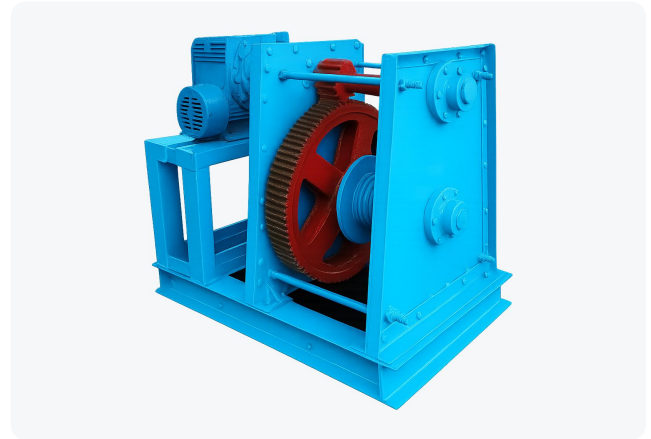
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# Door Lifting Power Winch

Coke Oven Gate Lifting Equipment

## AT A GLANCE

A power winch for lifting heavy coke oven doors. Its self-locking worm reducer gearbox holds the load steady even during a power failure. 5 - 7.5 HP motor, up to 5 tons.



## KEY SPECIFICATIONS



LIFTING CAPACITY  
**2.5 - 5 Tons**



MOTOR POWER  
**5 - 7.5 HP**



GEARBOX TYPE  
**Worm Reducer**



OPERATION  
**Vertical Lift**

## SPECIFICATIONS

|                  |                                        |
|------------------|----------------------------------------|
| Lifting Capacity | <b>2.5 - 5 Tons</b>                    |
| Motor            | <b>5 - 7.5 H.P. (3-Phase)</b>          |
| Operation        | <b>Vertical Lift</b>                   |
| Lifting Speed    | <b>Approx. 2 - 4 m / min</b>           |
| Gearbox          | <b>Self-Locking Worm Reducer</b>       |
| Drum             | <b>Grooved Steel Drum / Steel Drum</b> |

### DESIGN & DURABILITY

- ✓ Self-locking worm reducer gearbox
- ✓ Gears for a smooth, non-jerky lift
- ✓ Grooved steel drum or steel drum for wire rope

## BUYING INFORMATION

|                      |                                                                          |
|----------------------|--------------------------------------------------------------------------|
| <b>Lead time</b>     | Made to order. Ask us for the current lead time.                         |
| <b>Warranty</b>      | 1 year, as on all our machinery.                                         |
| <b>Installation</b>  | Installation supervision and commissioning available.                    |
| <b>Custom builds</b> | Capacity, motor power and dimensions matched to your plant and drawings. |

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## HOW IT WORKS

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### 01 Drive Activation

The electric motor engages and drives the worm gearbox input shaft through a coupling.

### 02 Torque Multiplication

The worm shaft drives the worm wheel, cutting speed and multiplying torque for a smooth, non-jerky lift.

### 03 Vertical Lift

The output shaft turns the grooved drum, winding the wire rope and lifting the oven door to the required height.

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## APPLICATIONS

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|                         |                                                                              |
|-------------------------|------------------------------------------------------------------------------|
| <b>Coke Oven Doors</b>  | Lifting and positioning heavy oven doors during charging.                    |
| <b>Heavy Dampers</b>    | Lifting isolation dampers in flue gas ducts in power plants and steel mills. |
| <b>Maintenance Bays</b> | General vertical lifting where overhead cranes are not available.            |

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## COMMON QUESTIONS

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### Why a worm gearbox instead of helical?

Worm gearboxes are self-locking: if power fails, the door's weight cannot drive the motor backwards.

### What is the lifting speed?

Slow and controlled, approximately 2 - 4 metres/minute, for safety.

### Does it come with a brake?

The worm reducer gearbox is self-locking, so the door's weight cannot drive the motor backwards if the power fails.

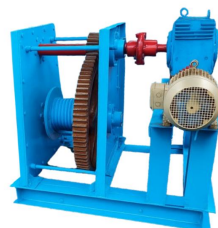
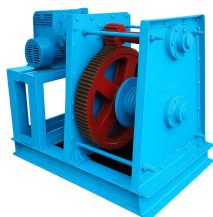
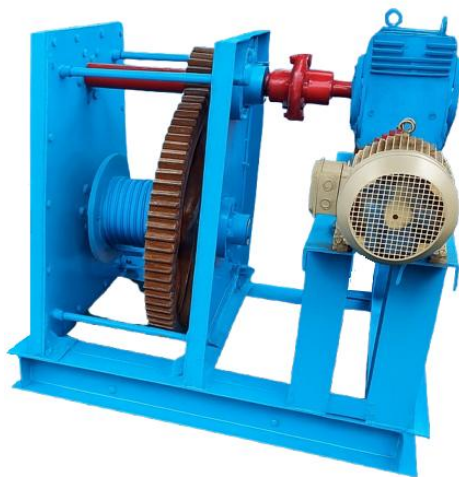
### What capacity do I need for a coke oven?

Standard coke oven doors usually need 2.5 to 5 tons, depending on battery height and door weight.

### Is the wire rope included?

Yes, a standard length is included; rope length and diameter can be customised.

## PRODUCT GALLERY



## ALSO FROM MANUAL TOOLS COMPANY

### Coal Crusher (5 No.)

Single Disc - Coal Disintegrator

### Coal Crusher (5 No.)

Double Disc - High Capacity Disintegrator

### Coke Cutter Machine

Double Drive, Drum Type - Industrial Series

### Ring Type Coke Cutter

Double Drive - Segmented Manganese Steel Rings

### Haulage Machine

10 HP / 10 Ton - Coke Oven Extraction Series

### Vibrator Screen Machine

Multi-Deck Series - Industrial Grading & Sorting

### Pusher Machine

With Roller Stamping Arrangement

### Coal Charging Car

Top Feed - Coke Oven Machinery Series

### Conveyor Components

Idlers, Rollers & Pulleys - Industrial Handling Systems

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